



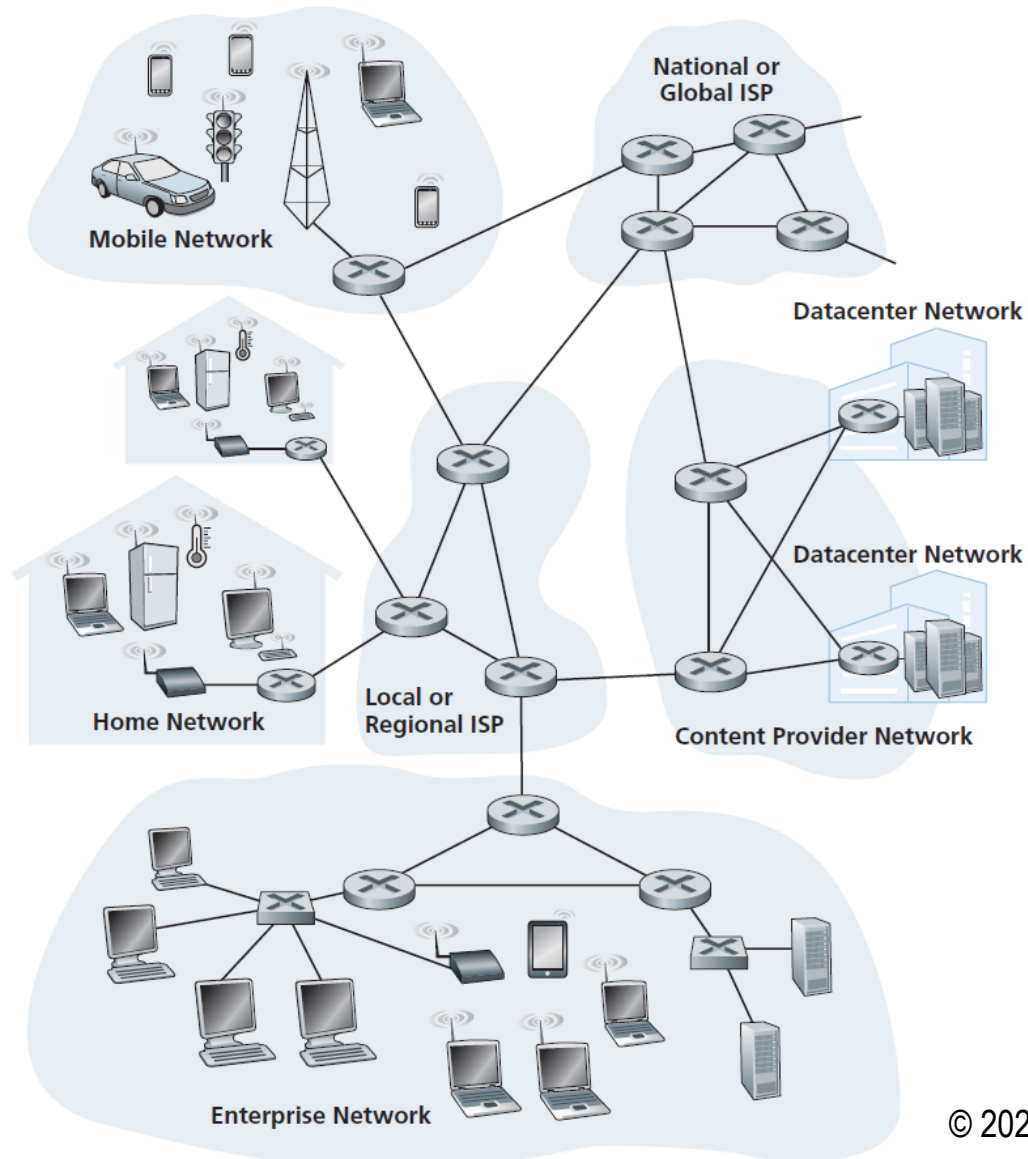
TCP/IP

SYSTEM INTEGRATION

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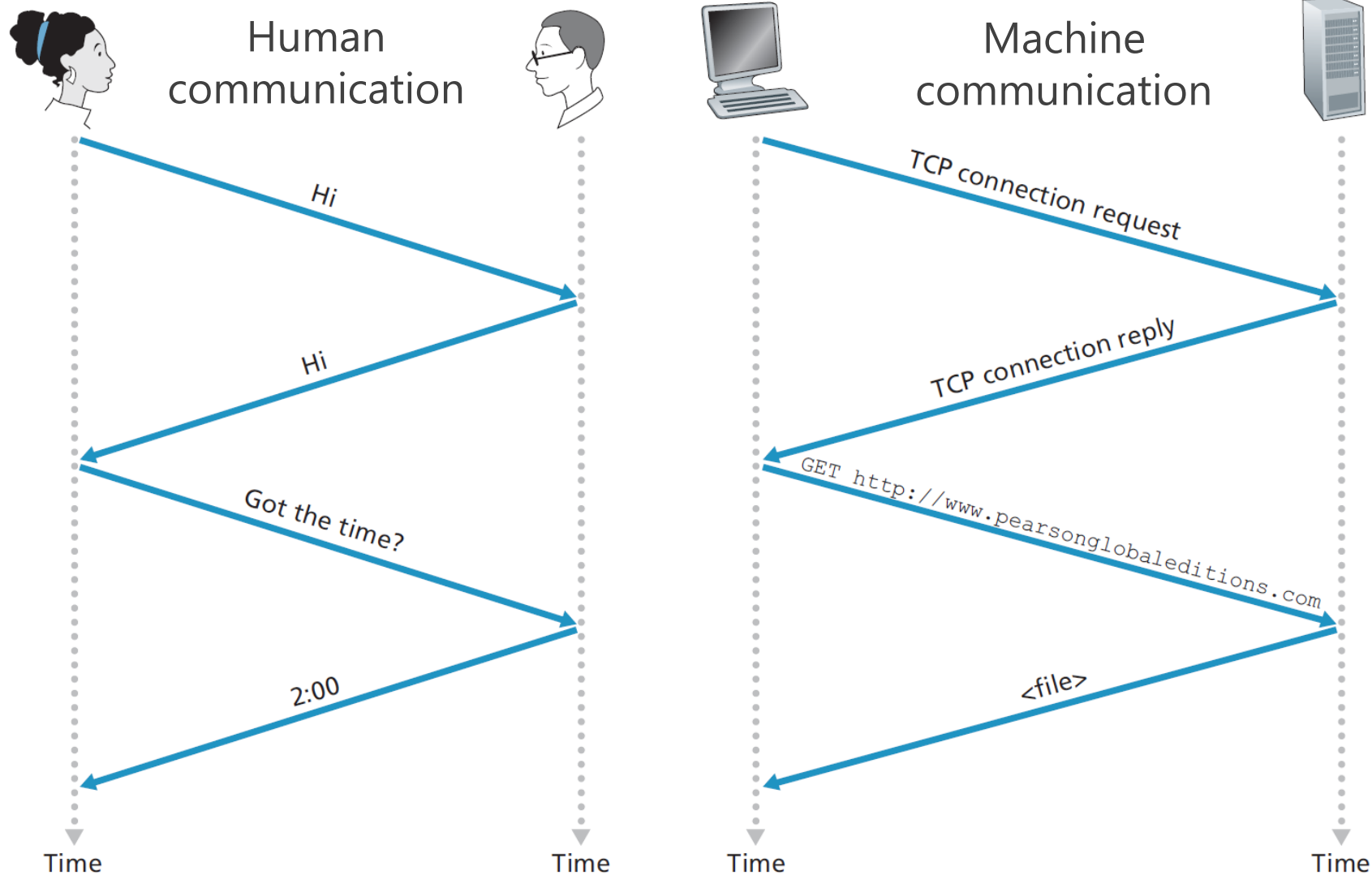
EK

The Internet



- Hosts
 - Servers
 - Clients
- Communication lines
- Communication devices
 - Routers
 - Switches
- ISPs (Internet Service Providers)
- The TCP/IP protocol

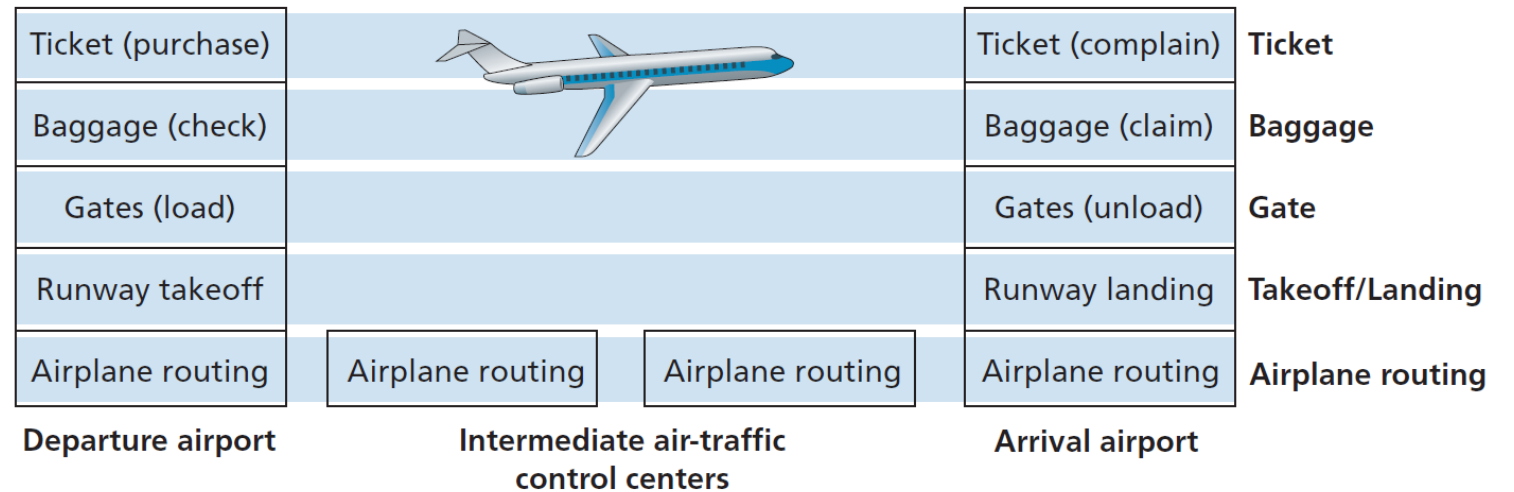
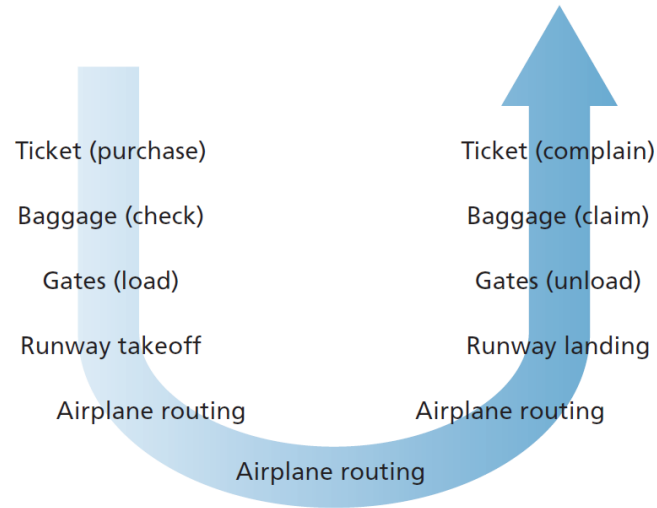
Protocols



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Protocol Layers

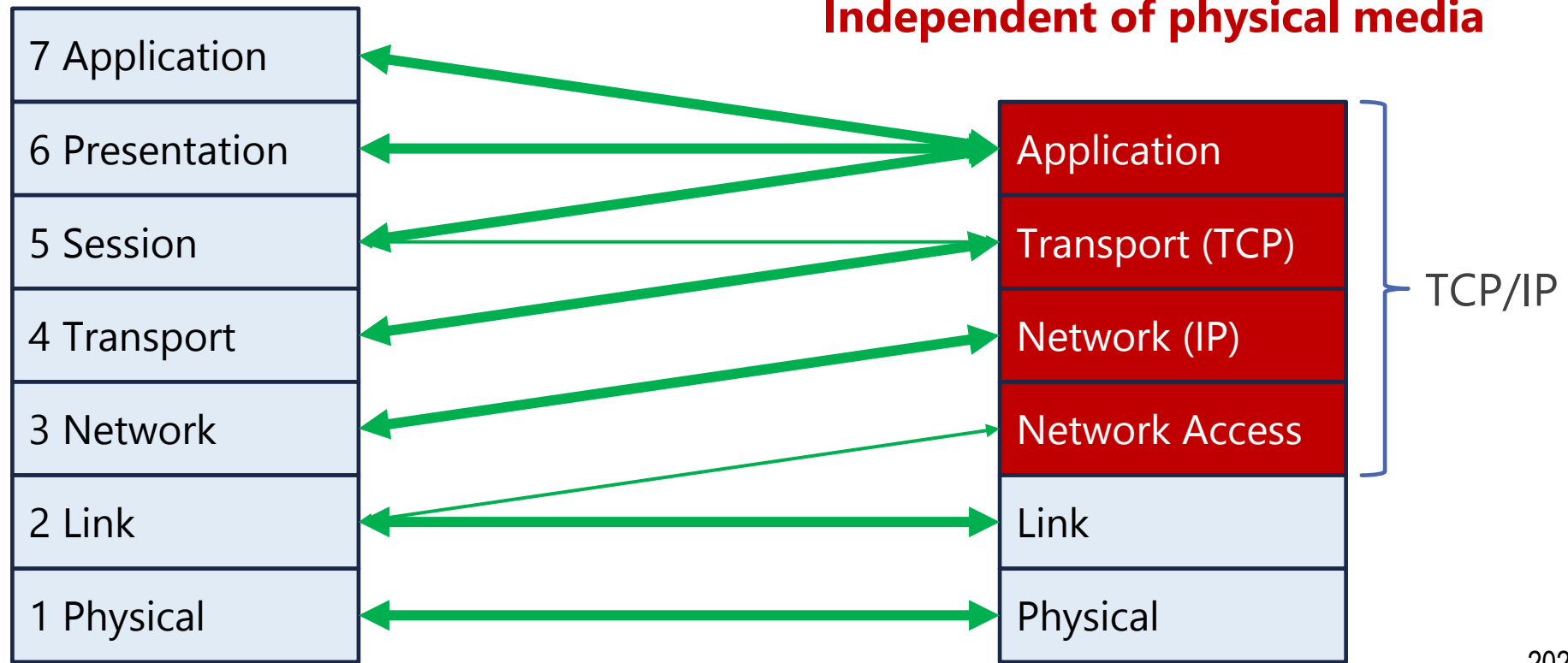
- Real-life analogy



OSI vs. TCP/IP

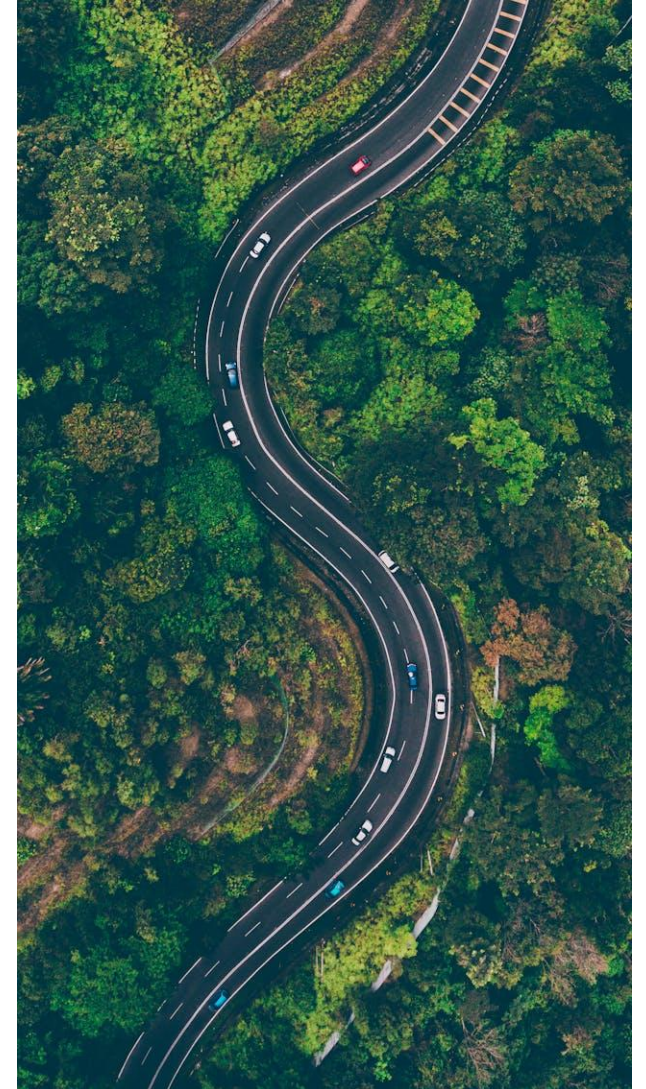
ISO OSI
(Open Systems Interconnection)
Reference model
7 layers

TCP/IP
(Transport Control Protocol/Internet Protocol)
Communications Protocol
4 layers



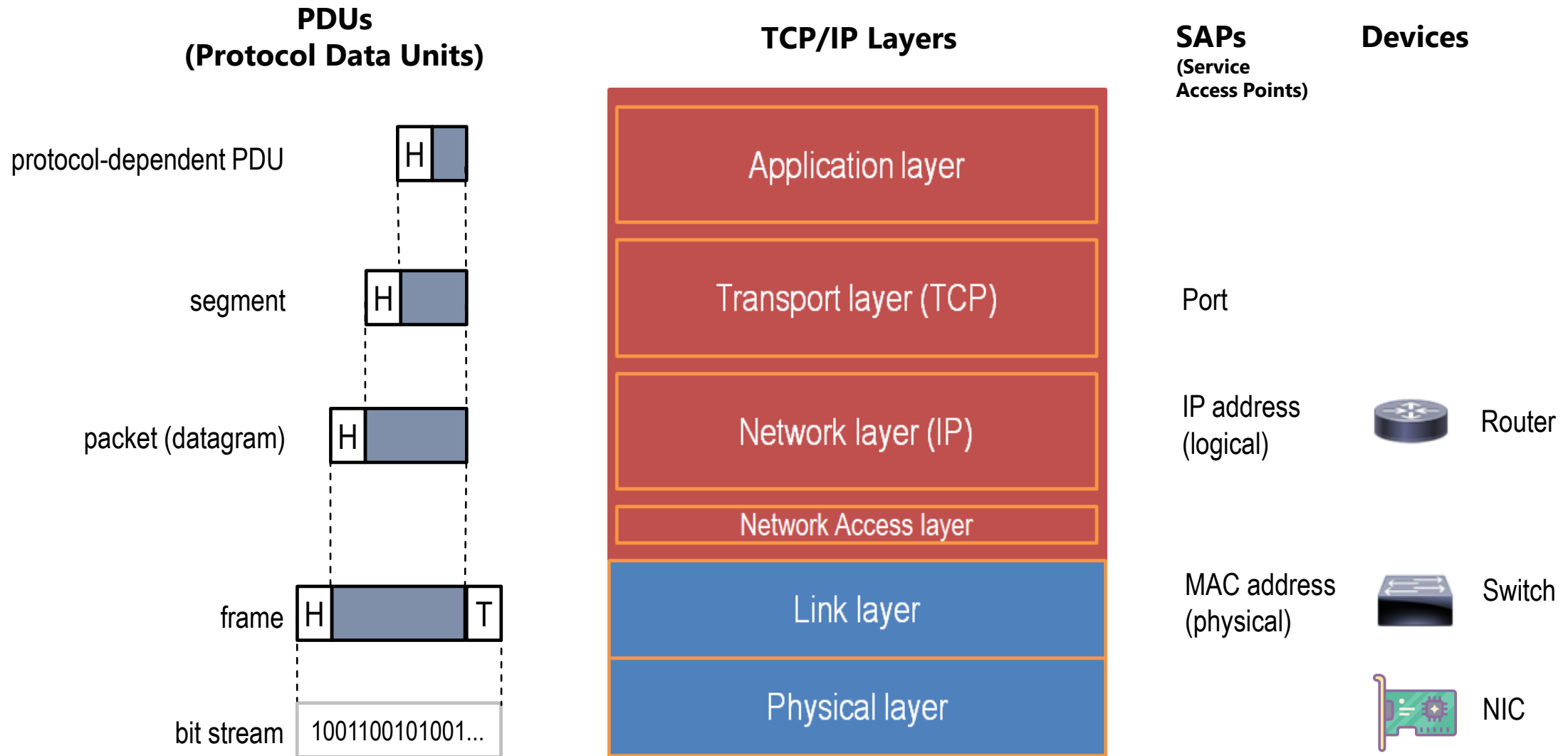
TCP/IP – Transport Control Protocol/Internet Protocol

- Actually a family of protocols
- Packet switching protocol
- Foundation of the Internet and most LANs and WLANs
- It groups networks together in a larger network
 - It communicates across networks, thus inter-net
- It allows for connection-establishing transmissions as well as for non-connection-establishing transmissions
- It lacks layers 1 and 2 of the OSI model, thus it can connect to
 - Any hardware
 - Any kind of physical and logical topology
 - Any data link layer protocol

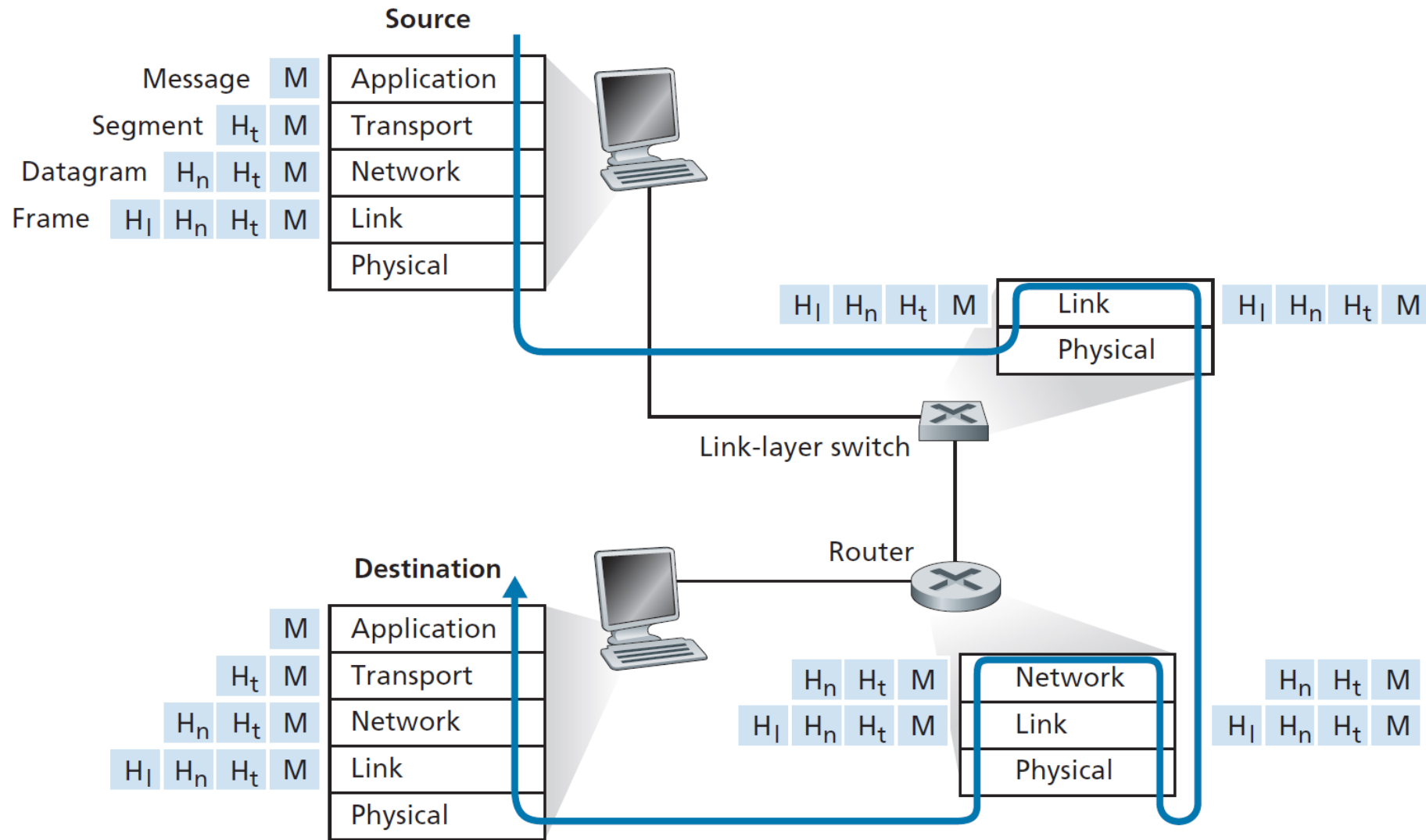


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TCP/IP Layers

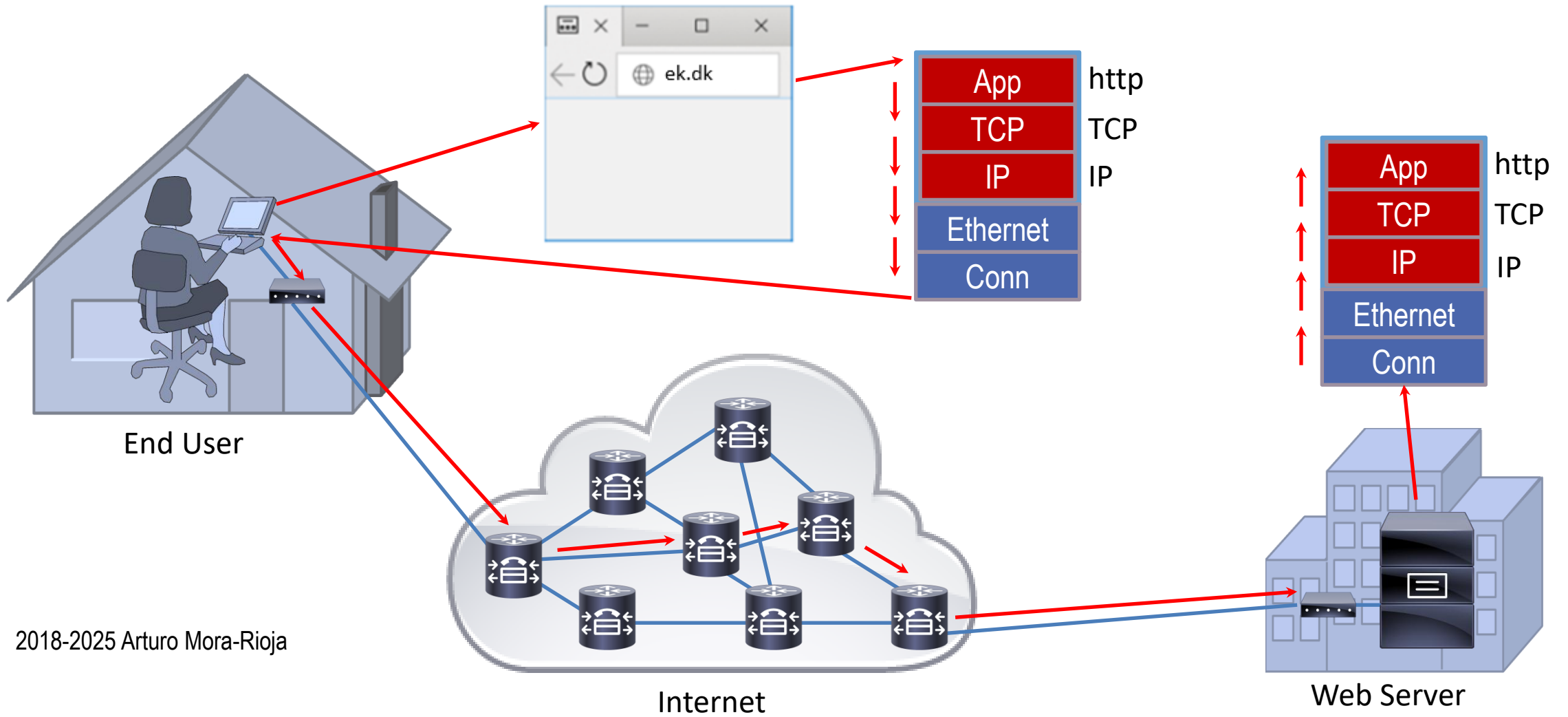


TCP/IP Layers



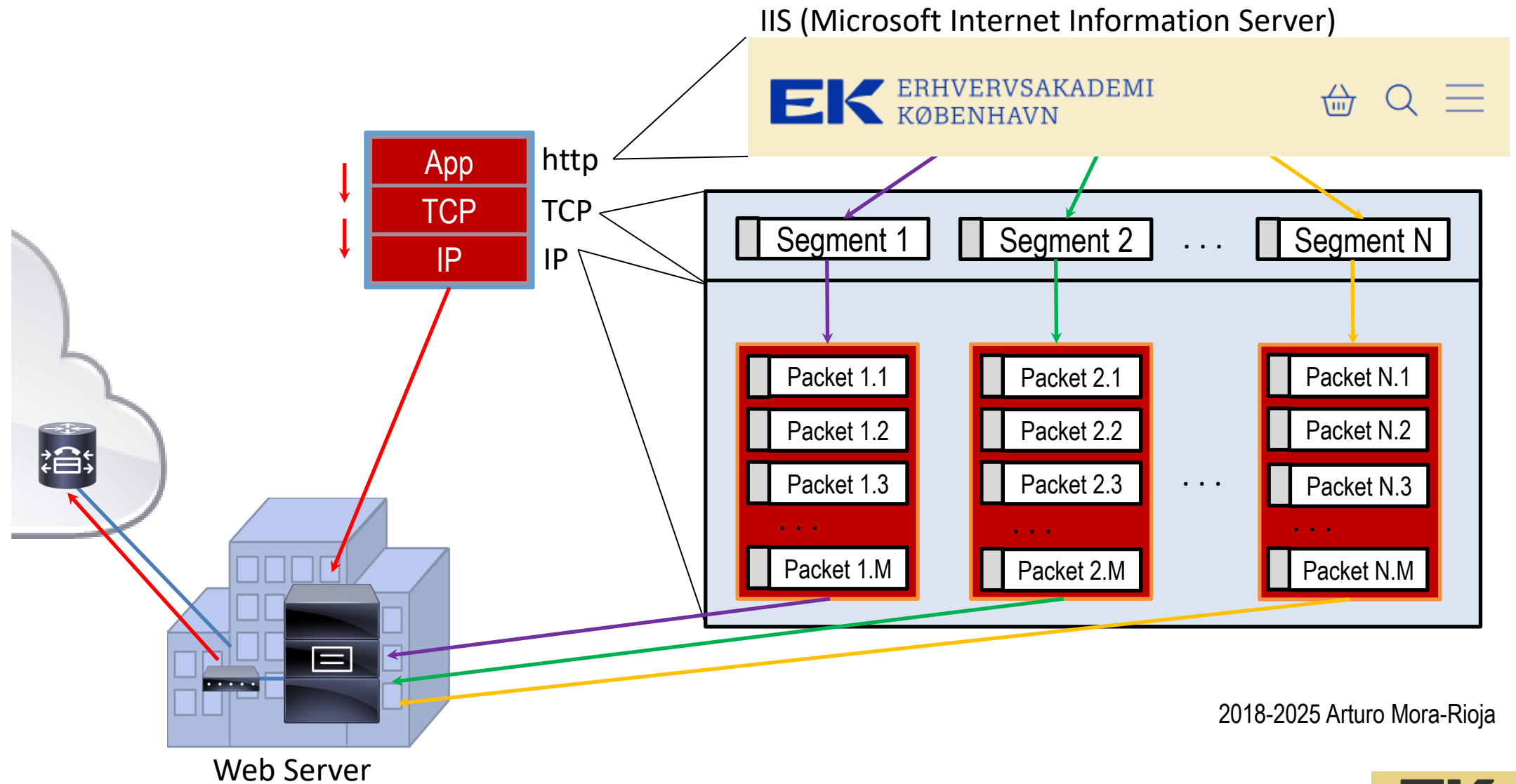
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TCP/IP Communication



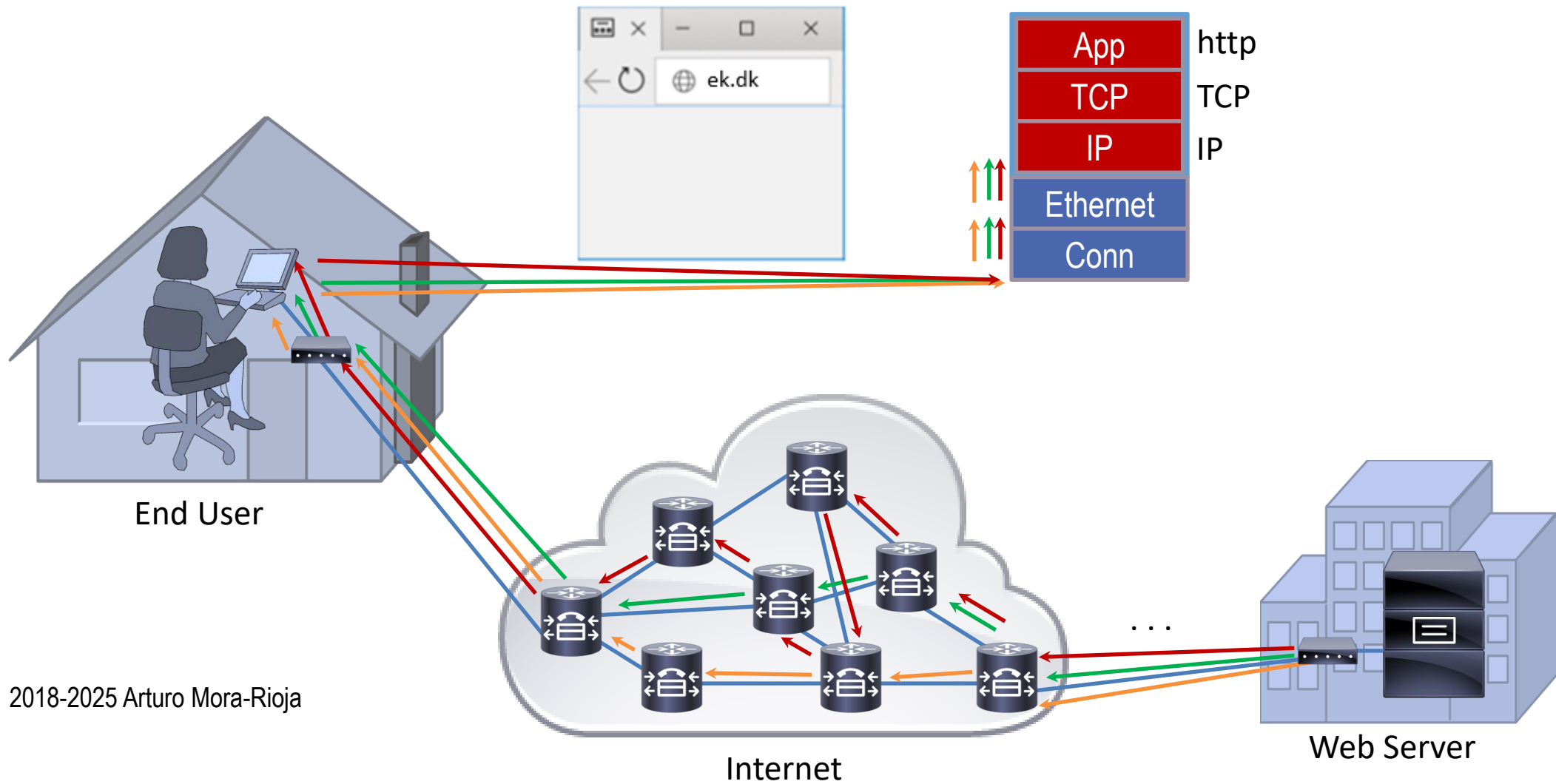
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TCP/IP Communication



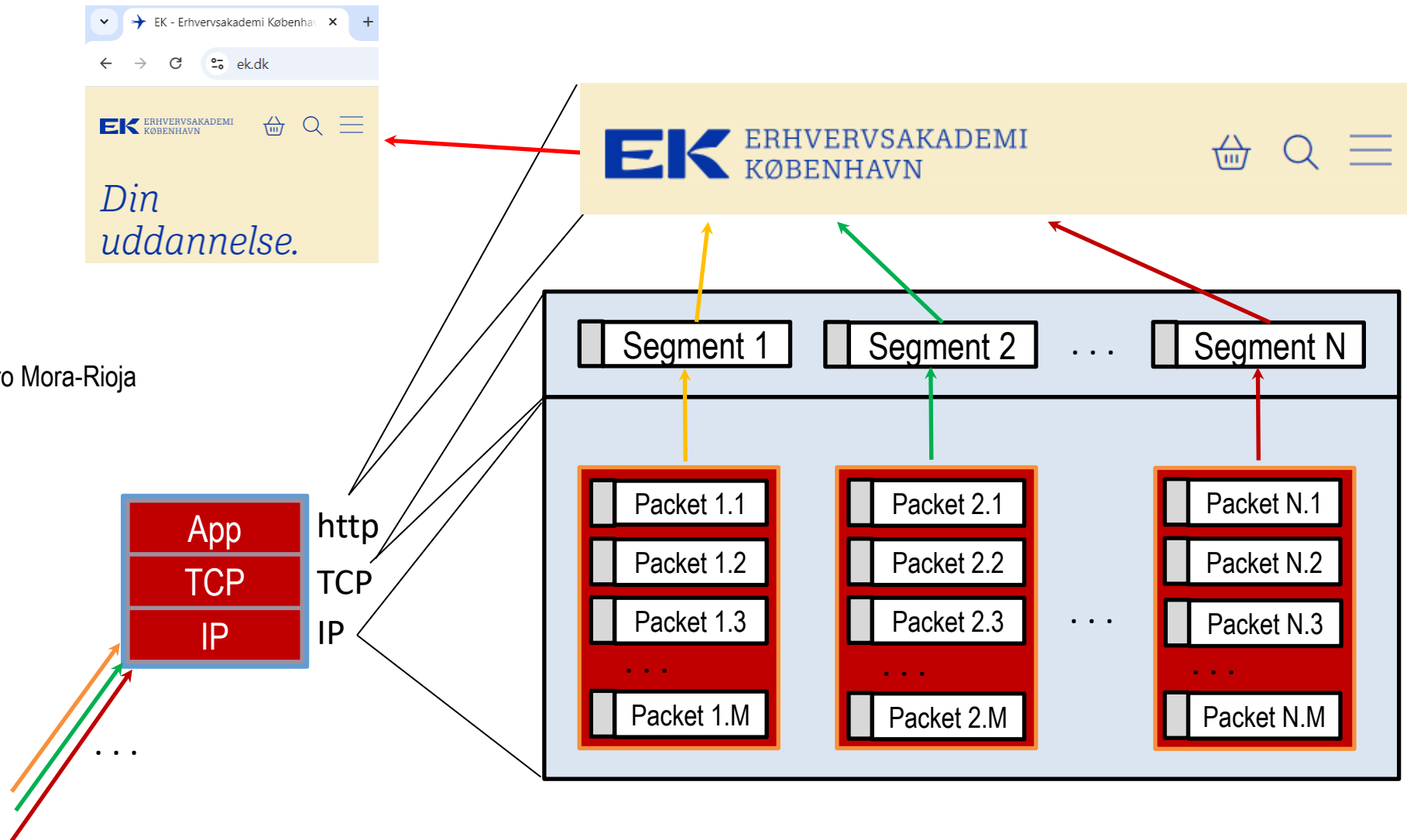
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TCP/IP Communication



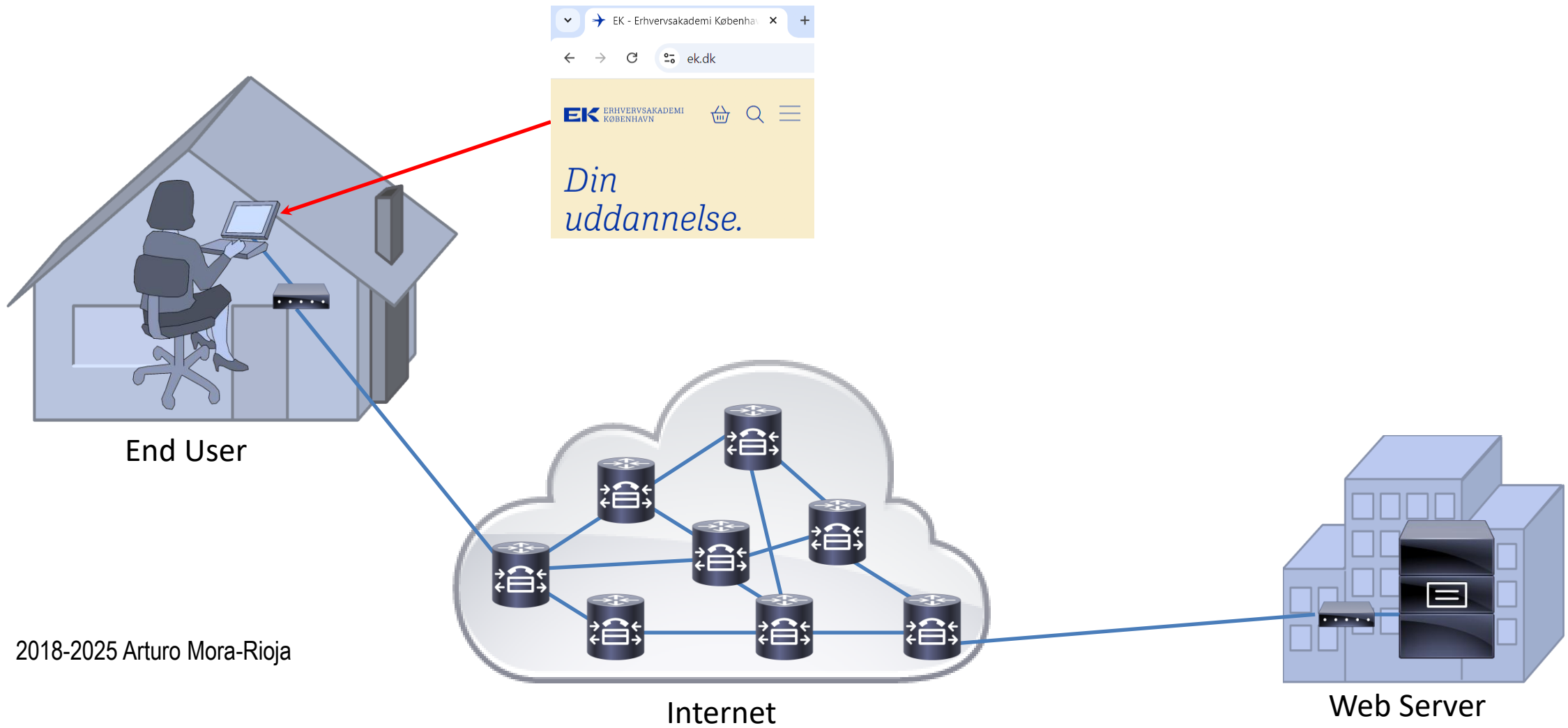
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TCP/IP Communication



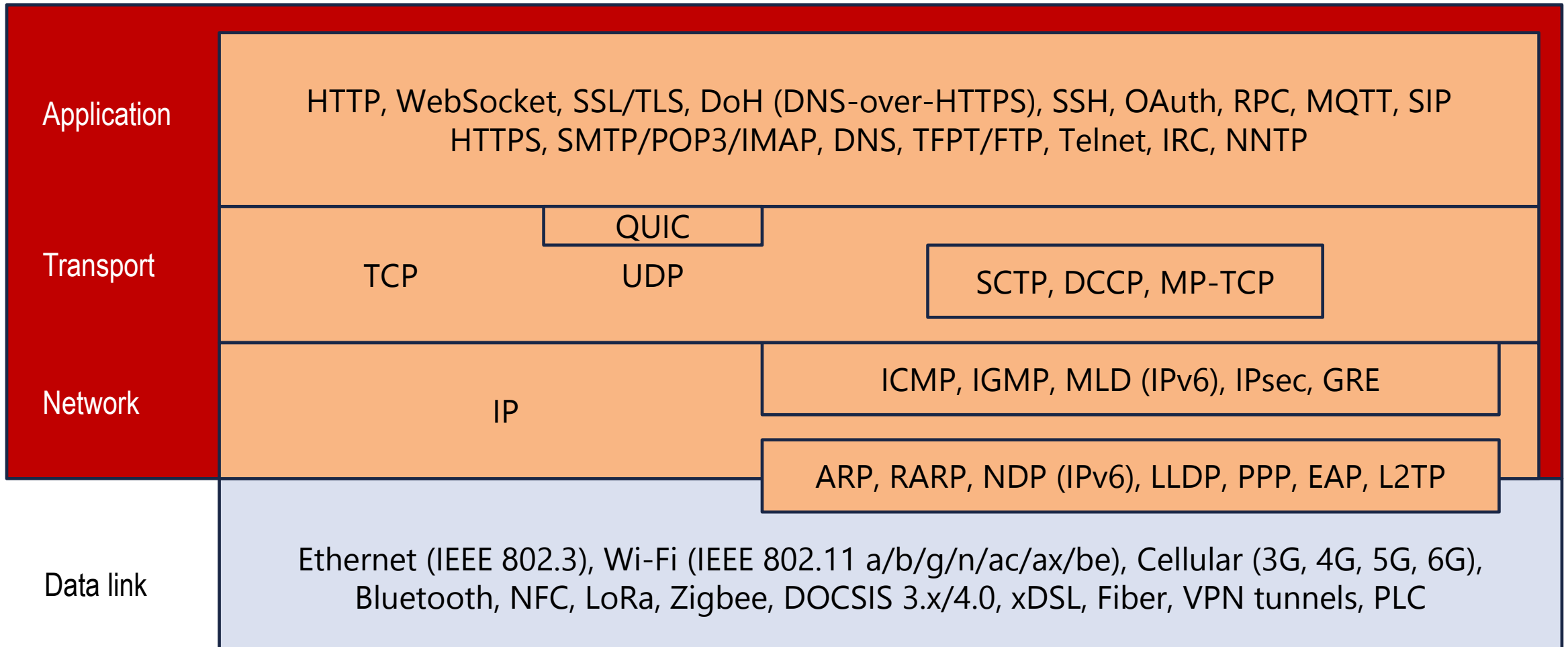
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TCP/IP Communication



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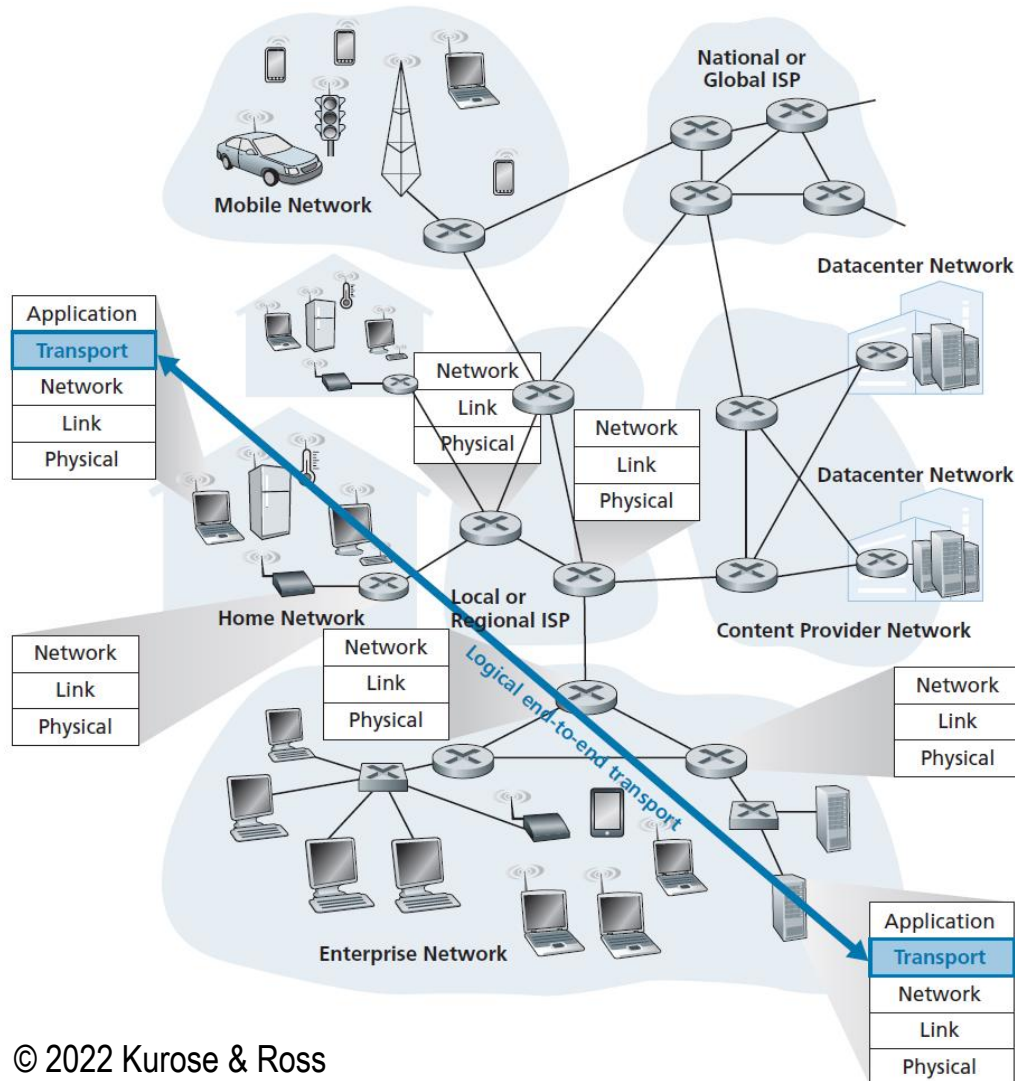
TCP/IP Protocols



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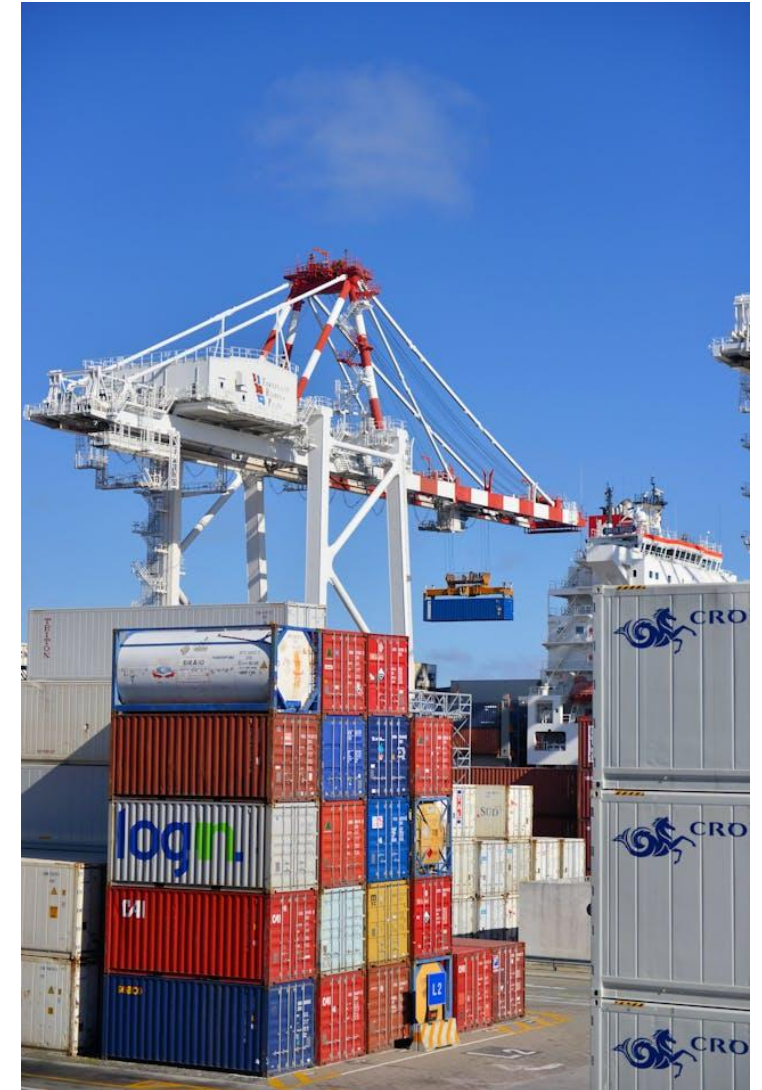
TCP/IP: The Transport Layer

- End-to-end logical communication
- It can guarantee the ordered reception of information



Transport Ports

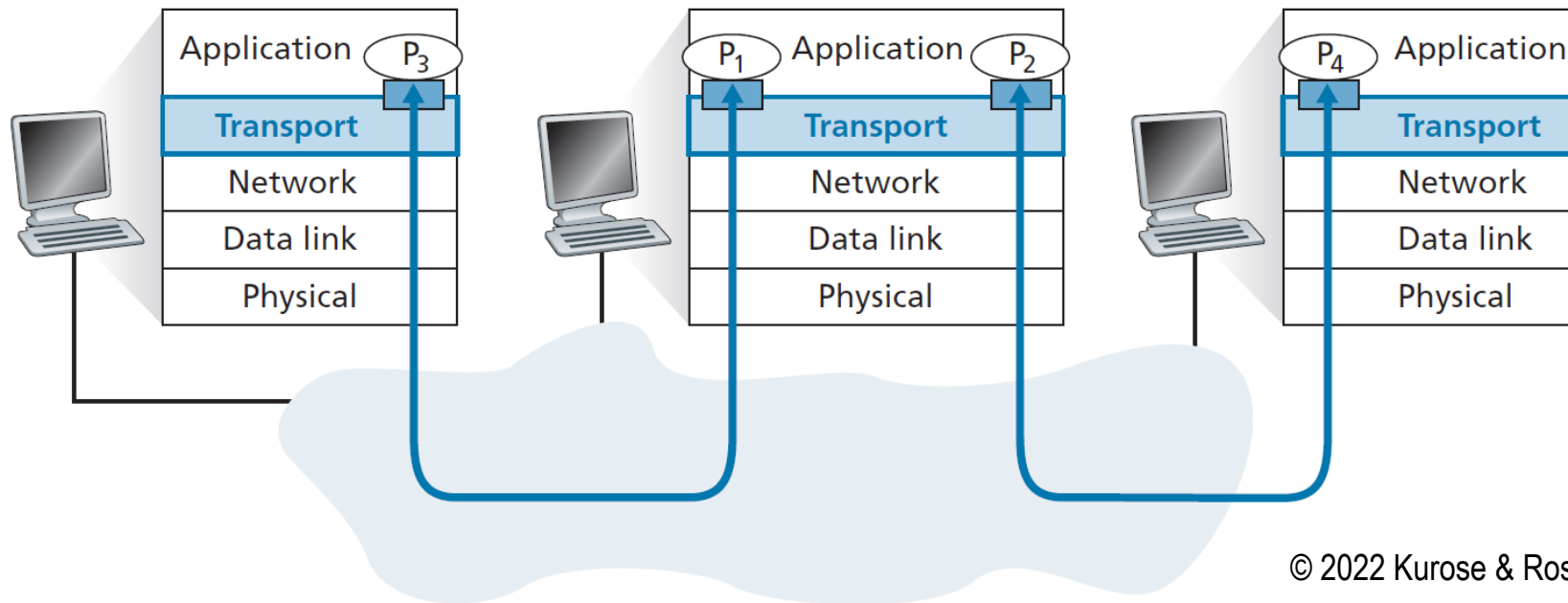
- **Port:** Transport-layer address (service access point)
- Represented by 16-bit numbers (0-65535)
- Three categories:
 - System ports = well-known ports (0 to 1023)
 - Used by standard Internet services
 - Registered ports = user ports (1024 to 49151)
 - Dynamic ports = private ports (49152 to 65535)
- System and registered ports require registration at IANA (Internet Assigned Numbers Authority)
 - [List of system and registered ports](#)
- **Socket:** IP address + port number (e.g., 20.50.2.66:443, 127.0.0.1:8080)



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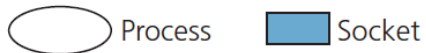
Multiplexing and Demultiplexing

- Sockets communicate with application-level processes
- **Multiplexing:** Encapsulating transport-level information into a segment
- **Demultiplexing:** Forwarding a segment to the correct socket



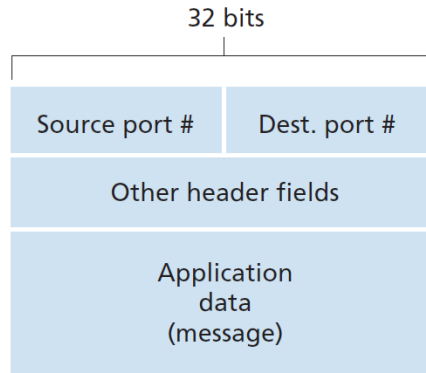
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Key:

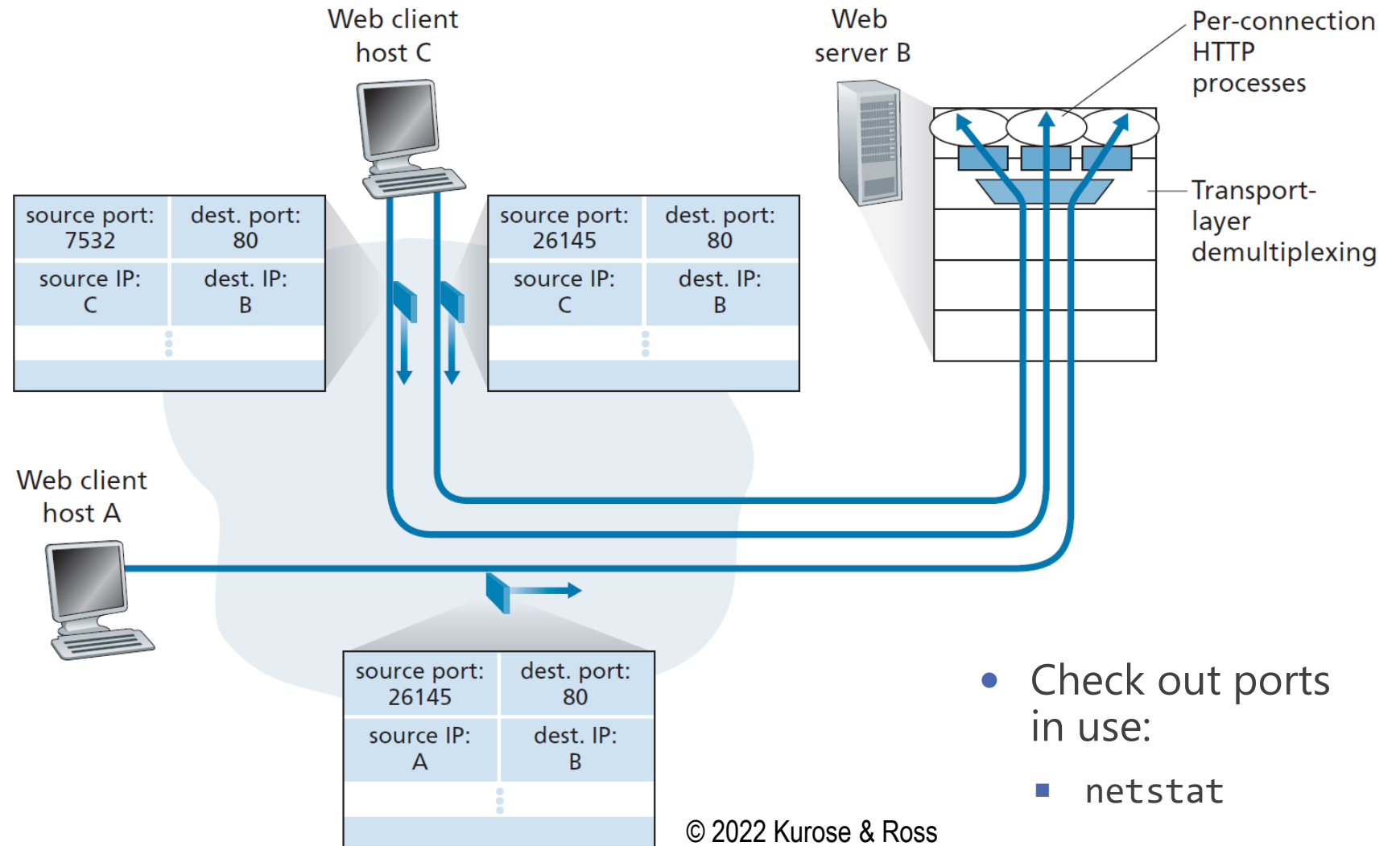


Multiplexing and Demultiplexing

- Transport-layer segment:



- A server can process multiple requests even from the same client
- Persistent HTTP connections reuse the same socket

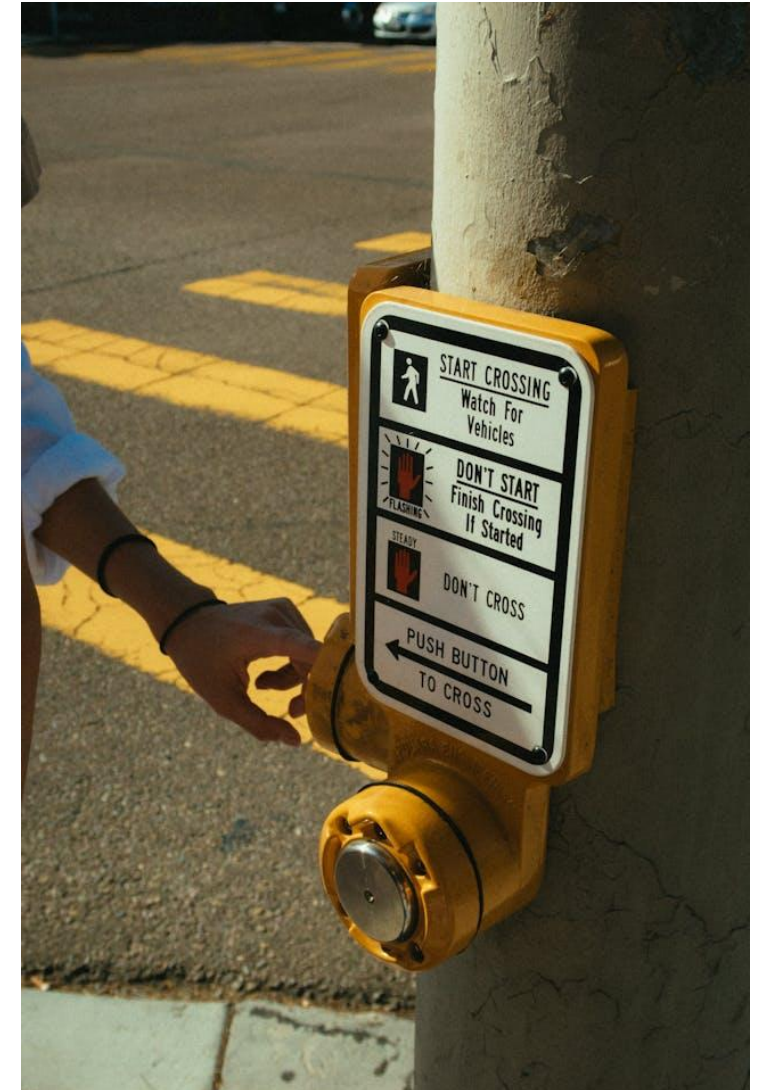


- Check out ports in use:
 - netstat

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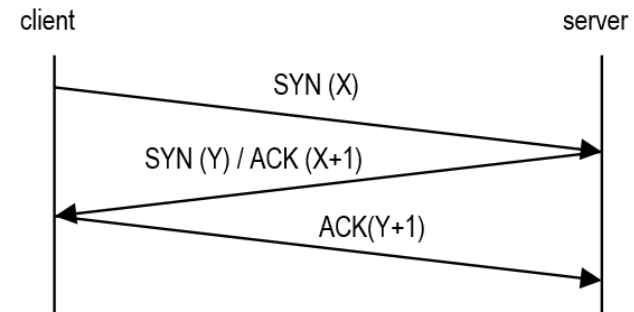
TCP – Transport Control Protocol

- Connection-oriented
- Reliable
 - Use of acknowledgements
 - Segment order control
- It segments the information for its transmission, which it reassembles in the destination
- Flow control
 - It matches the speeds of the sender and the receiver to prevent flooding
 - It also supports pipelining: the client does not wait for each segment's acknowledgement

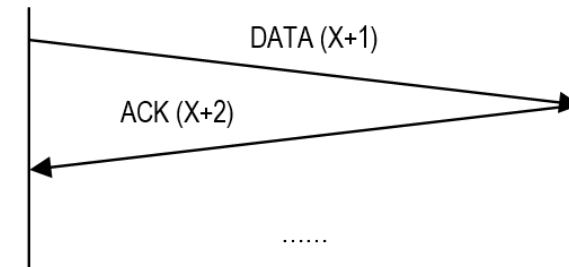


TCP Connection

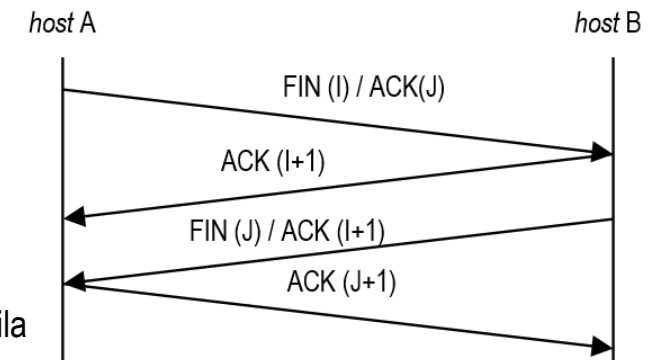
- Connection (3-way handshake)



- Data interchange

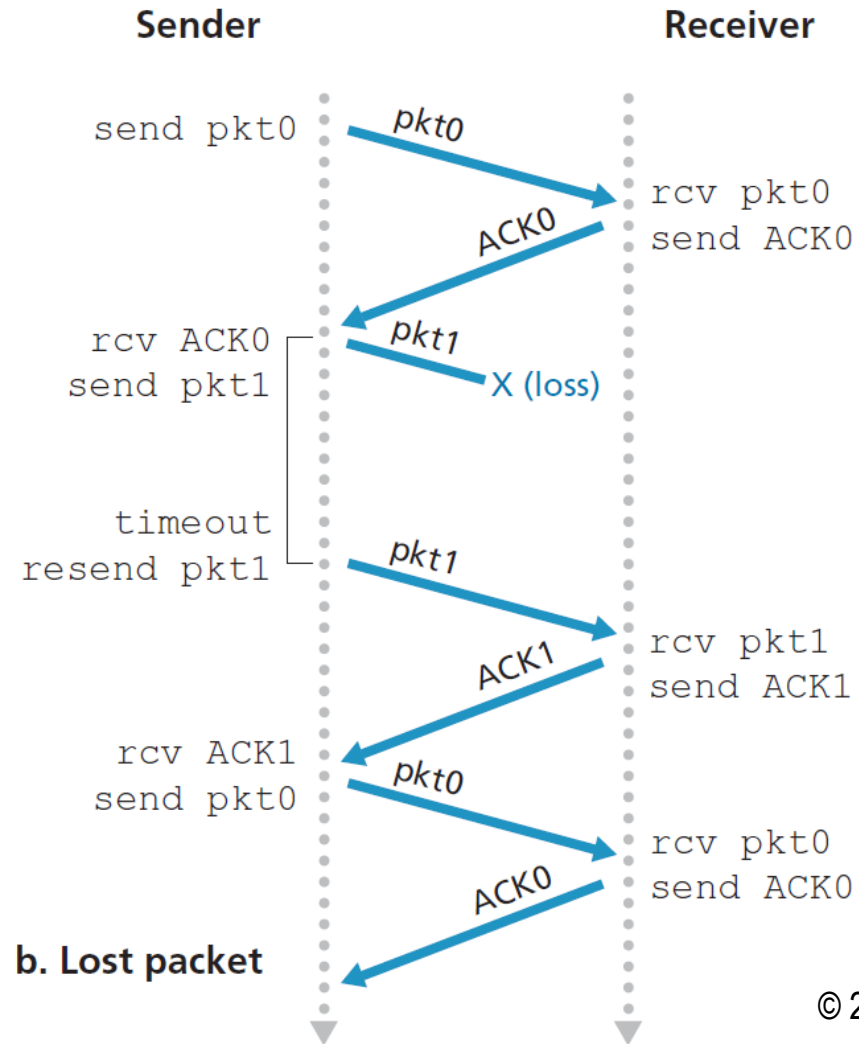
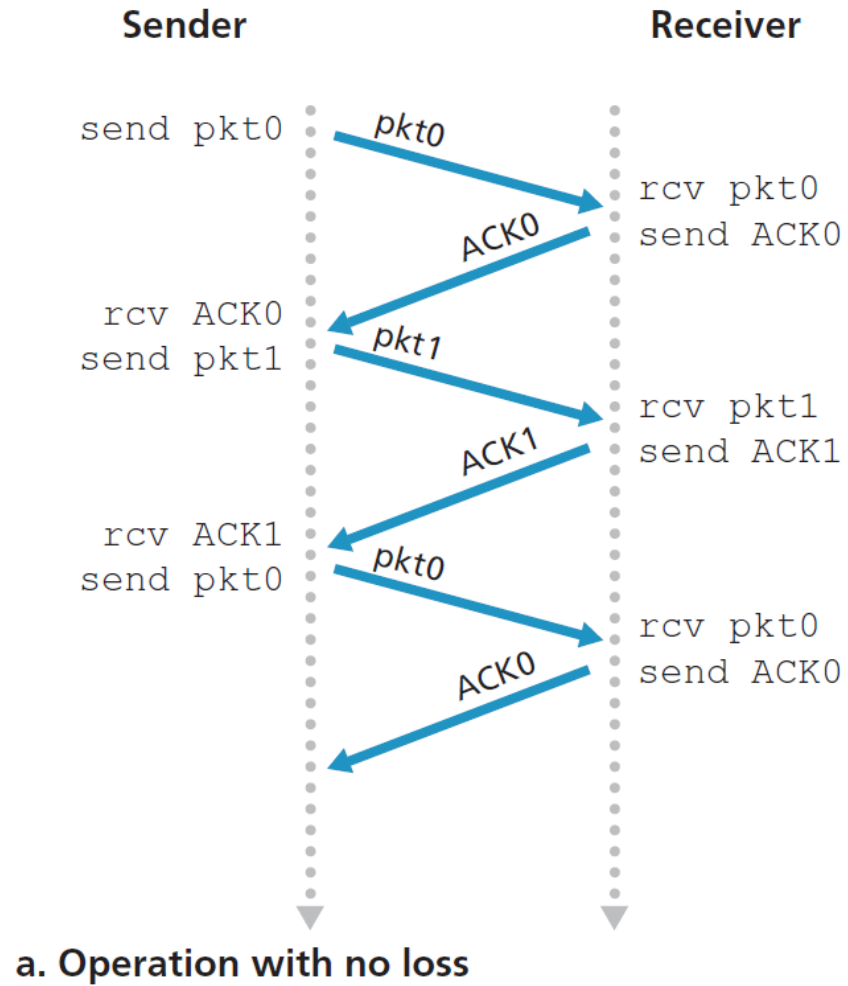


- Disconnection (4-way handshake)



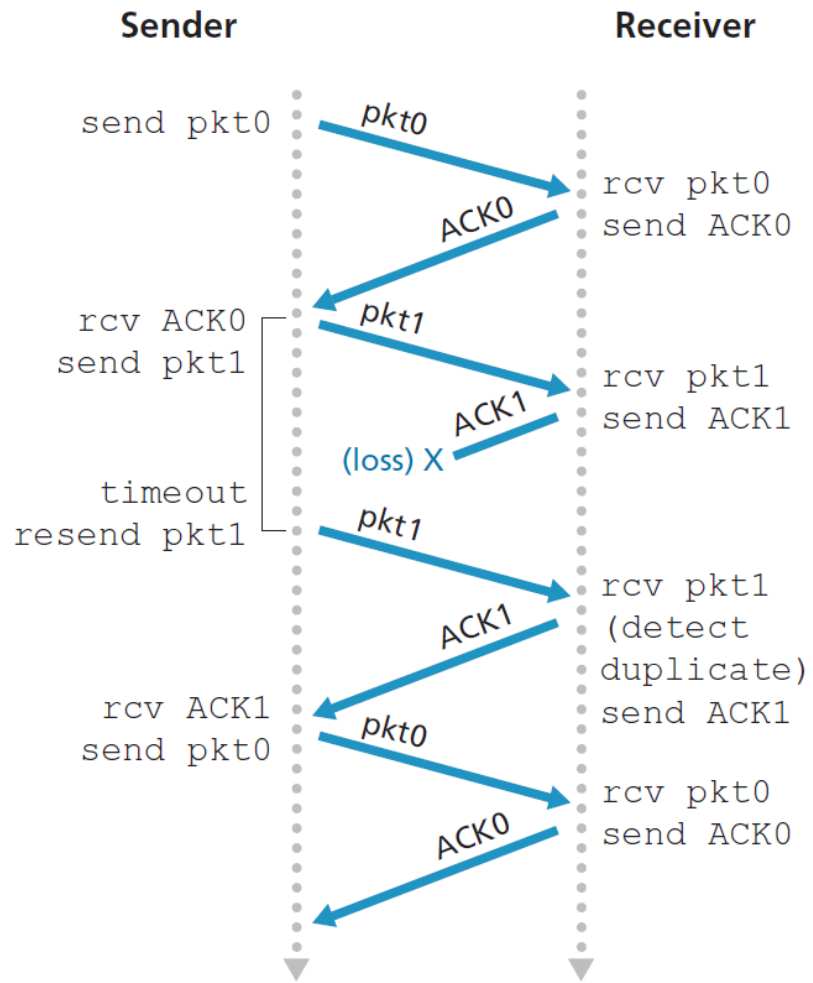
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Reliability in TCP

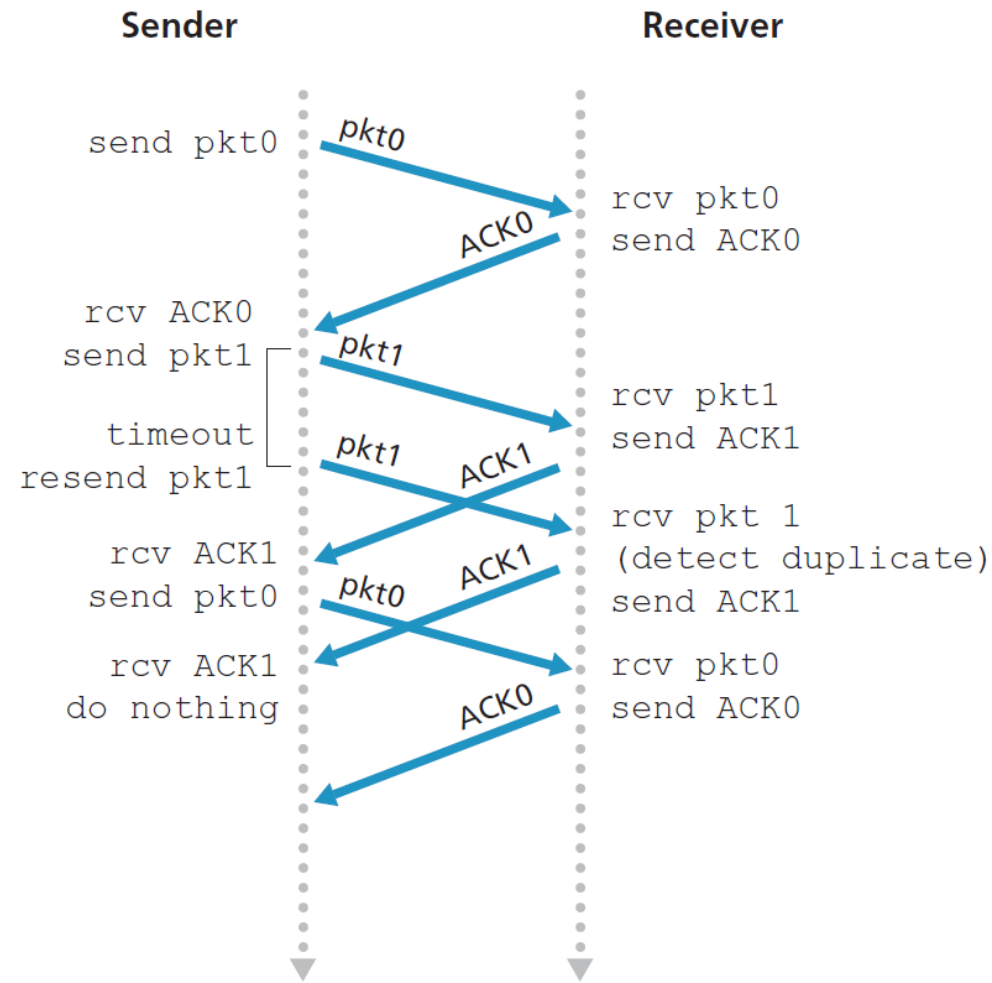


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Reliability in TCP

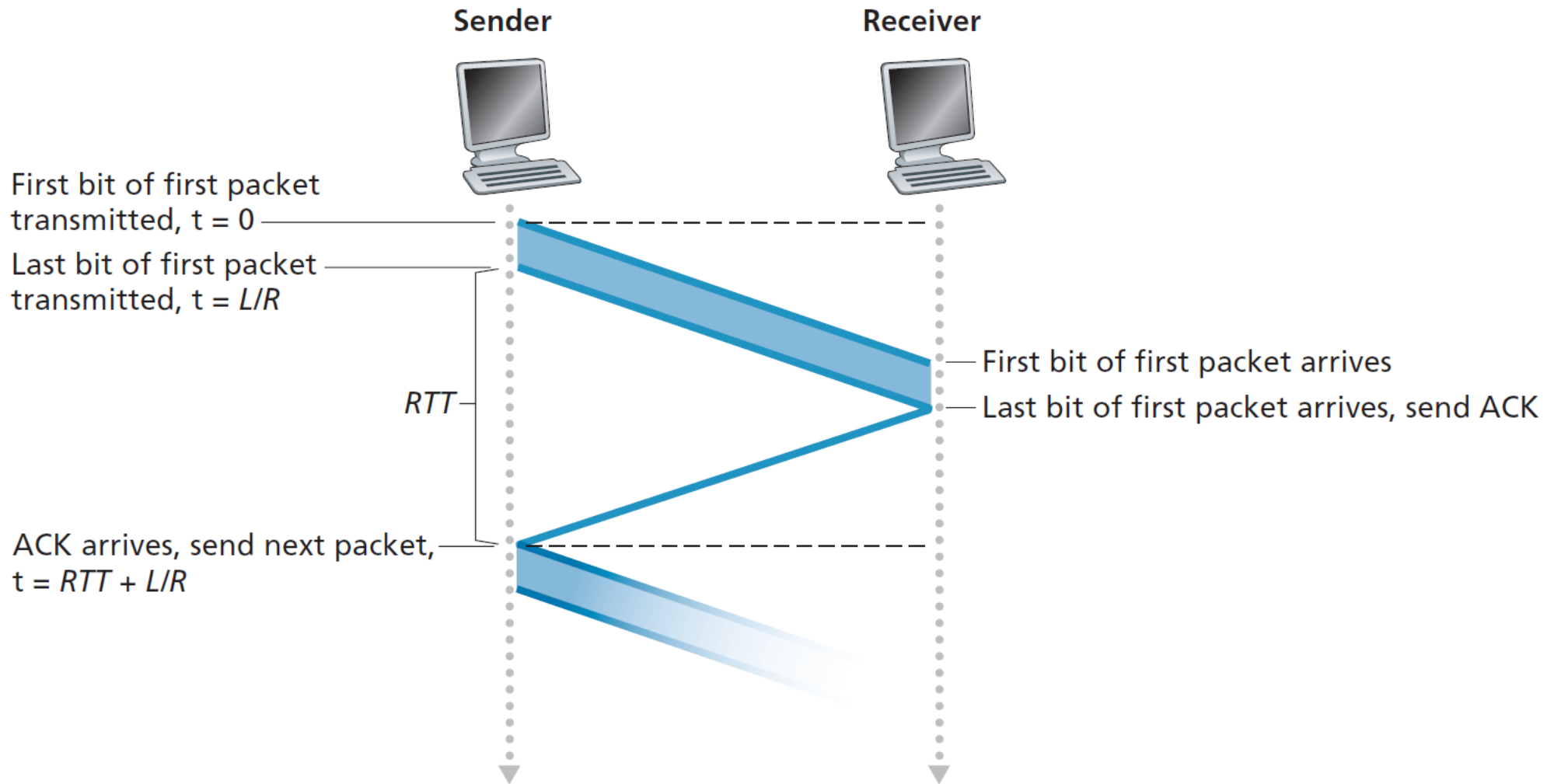


c. Lost ACK



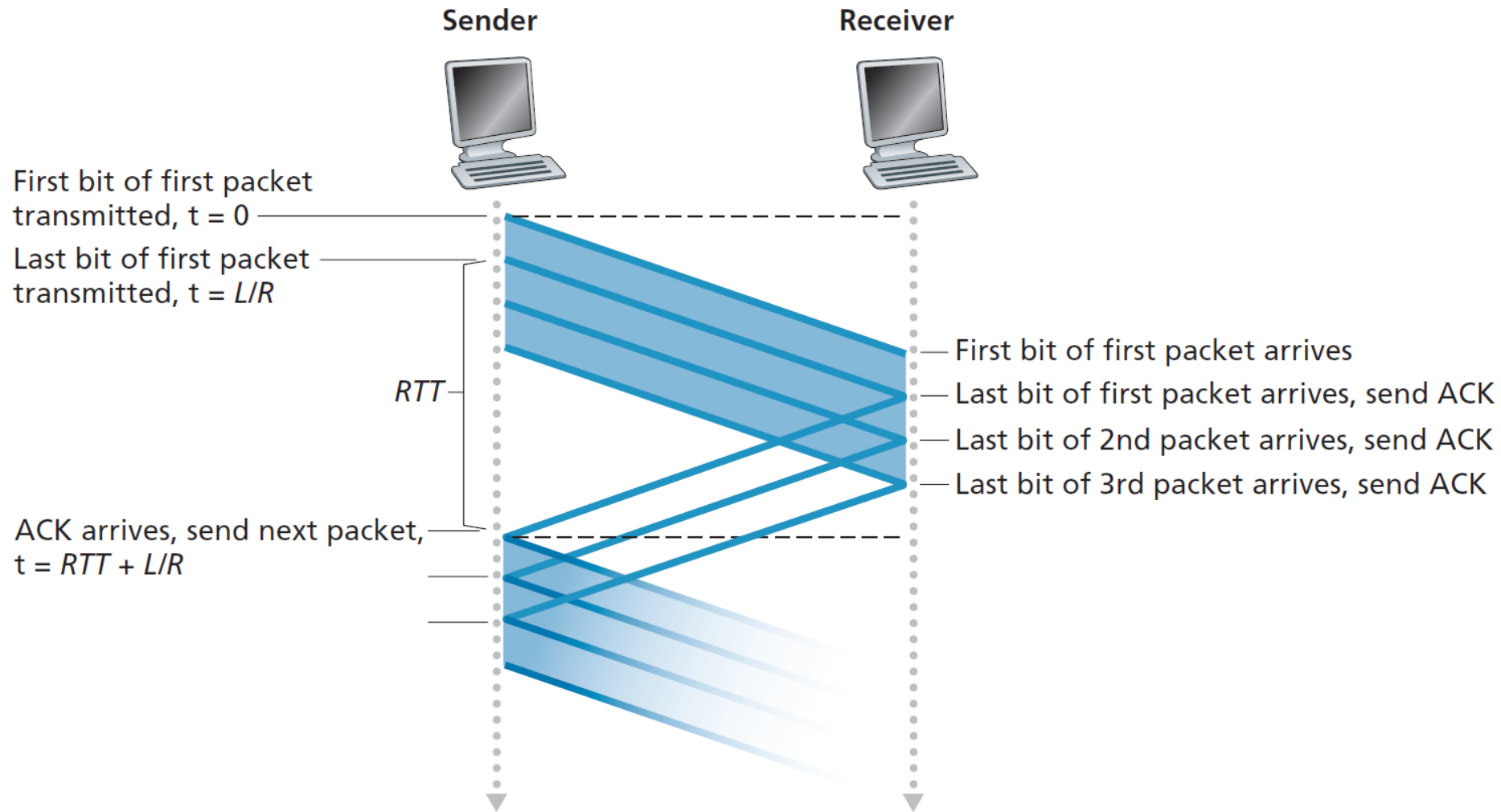
d. Premature timeout

Pipelining



a. Stop-and-wait operation

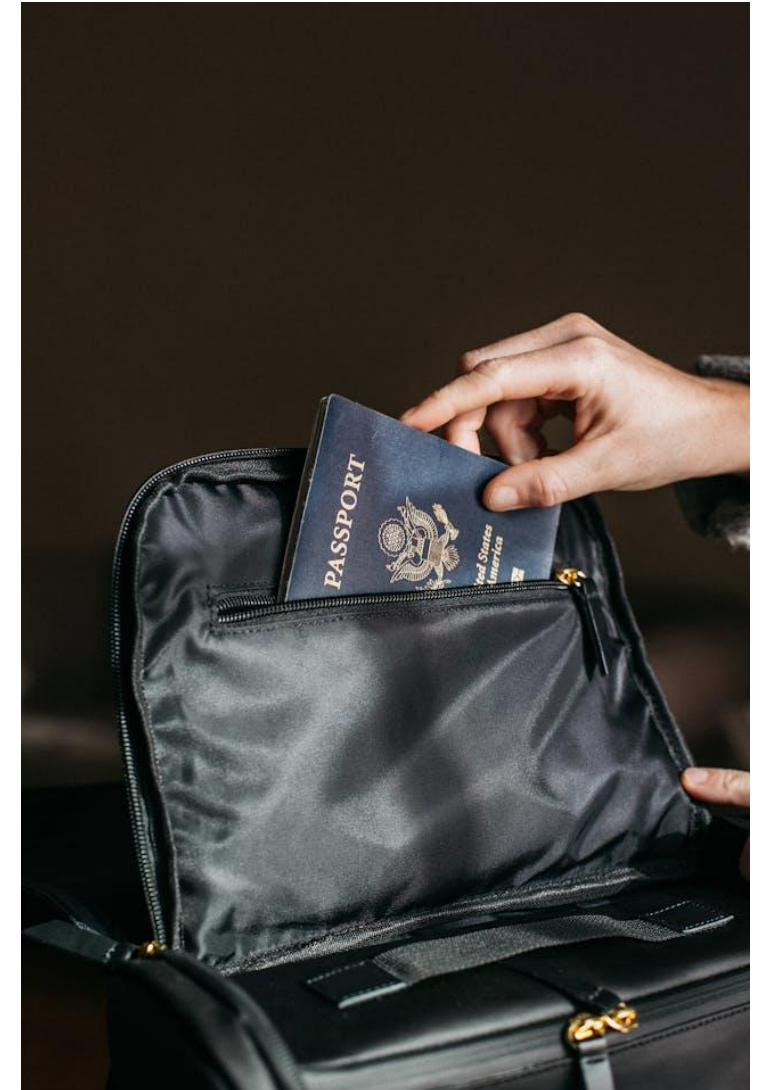
Pipelining



b. Pipelined operation

SSL/TLS – Secure Socket Layer/Transport Layer Security

- Protocol living between the application and transport layers
- Provides security for TCP connections at transport level
- Application-specific
 - HTTP → HTTPS
 - FTP → FTPS
 - Etc.
- The acronyms SSL and TLS are used almost interchangeably, but SSL is long deprecated
- It uses digital certificates
 - Encryption → confidentiality
 - Digital signatures and hashing → authentication, integrity, non-repudiation



UDP – User Datagram Protocol

- Connectionless
- Unreliable
- Fast
 - Lack of acknowledgements
 - Lack of sorting mechanisms
 - No delay due to connection establishment
 - No delay due to congestion control
 - Small packet header overhead (8 bytes vs. TCP's 20 bytes)
- Used for real-time services (e.g., streaming)

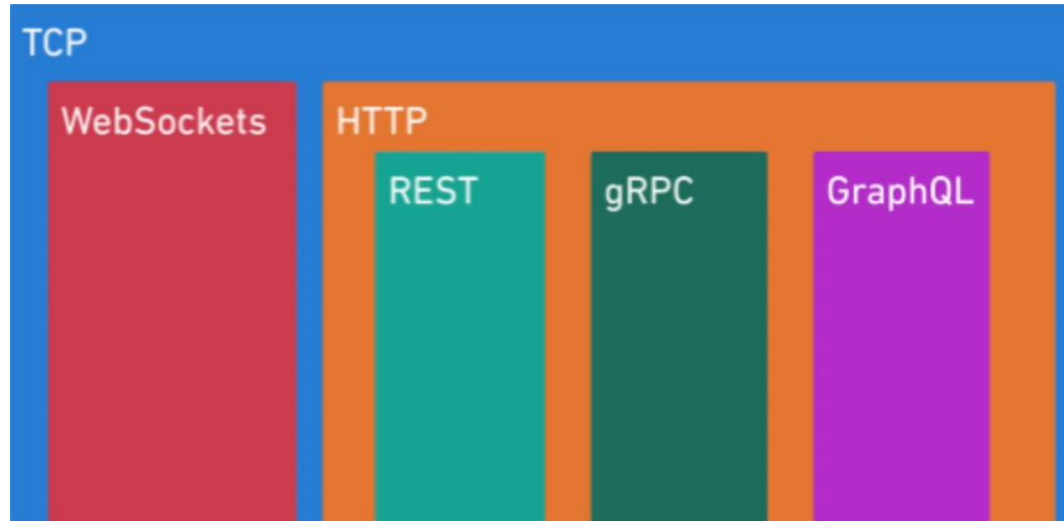


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TCP vs. UDP

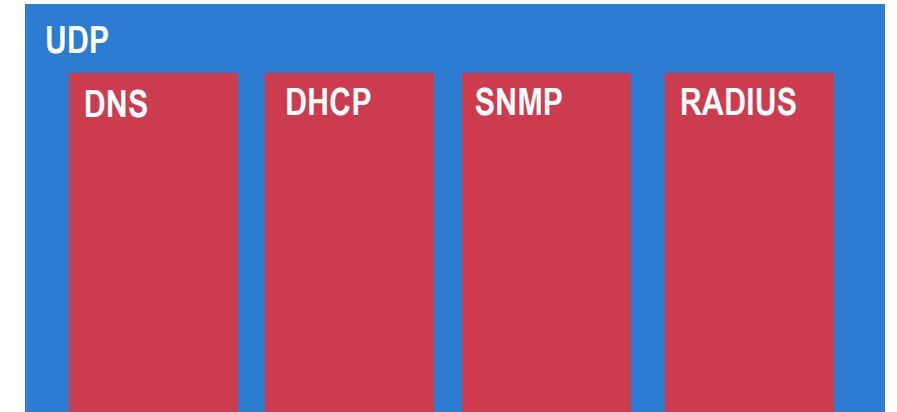
- TCP

- 👍 Reliable
- 👍 Ordered
- 👍 Error-checked
- 🙅 Slower than other protocols



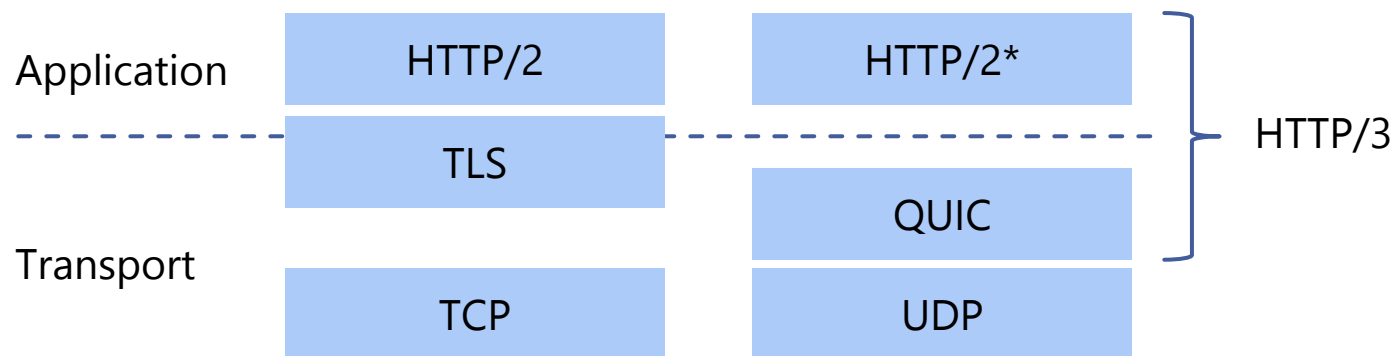
- UDP

- 👍 Fast
- 🙅 Unreliable
- 🙅 Unordered
- 🙅 No error checks



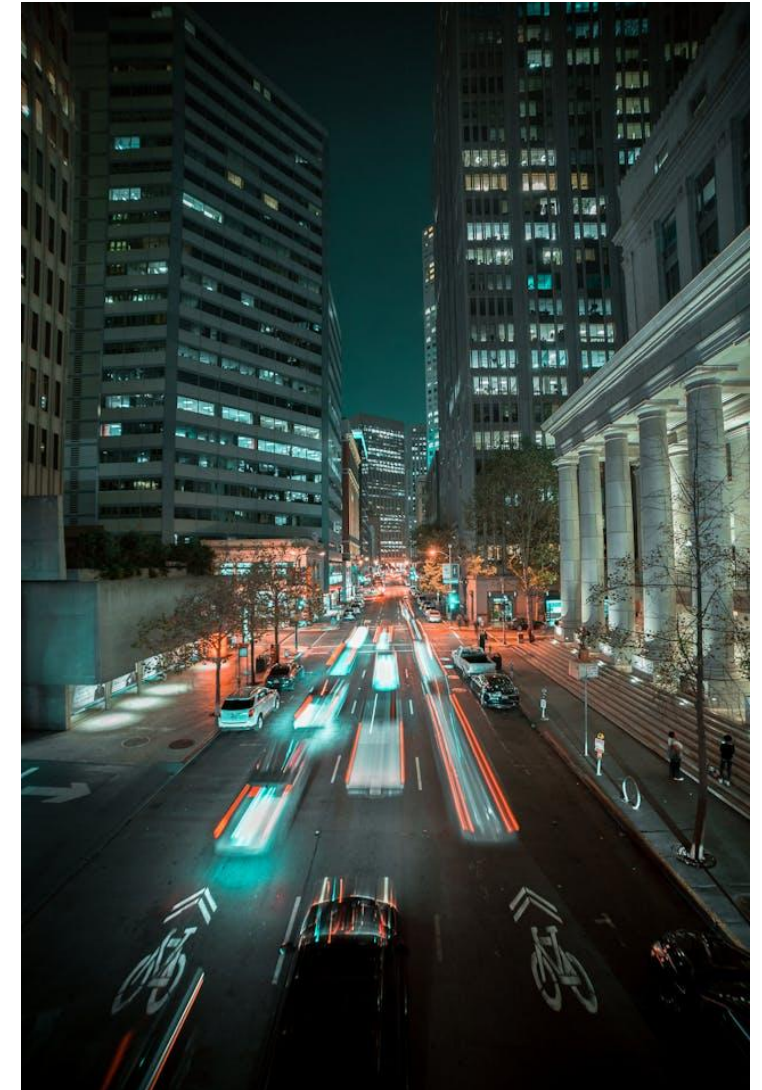
QUIC – Quick UDP Internet Connection

- Substitutes TCP and TLS
 - It implements reliability in the transport layer
 - Then runs on top of UDP
 - Secure. Mandatory encryption
- Implemented by Google
- Used by 8,6% of all websites, but 75% of Meta's traffic
- Base protocol for HTTP/3



* Only the parts not covered by QUIC

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Usage of Transport Layer Protocols by Application

Application	Application-Layer Protocol	Underlying Transport Protocol
Electronic mail	SMTP	TCP
Remote terminal access	Telnet	TCP
Secure remote terminal access	SSH	TCP
Web	HTTP, HTTP/3	TCP (for HTTP), UDP (for HTTP/3)
File transfer	FTP	TCP
Remote file server	NFS	Typically UDP
Streaming multimedia	DASH	TCP
Internet telephony	typically proprietary	UDP or TCP
Network management	SNMP	Typically UDP
Name translation	DNS	Typically UDP

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Port to Protocol Association

Port	TCP	UDP	Protocol / Description
20	✓		FTP (data transfer)
21	✓		FTP (control connection)
22	✓		SSH (Secure Shell, SFTP, SCP)
25	✓		SMTP (email delivery)
53	✓	✓	DNS (Domain Name System)
67		✓	DHCPv4 client
68		✓	DHCPv4 server
69		✓	TFTP (Trivial File Transfer Protocol)
80	✓	✓	HTTP (web traffic)
110	✓		POP3 (email retrieval)
123		✓	NTP (time synchronization)
143	✓		IMAP4 (email retrieval)
161		✓	SNMP (monitoring)
162	✓	✓	SNMP Trap (notifications)
179	✓		BGP (routing protocol)
389	✓	✓	LDAP (directory services)
443	✓	✓	HTTPS (secure web traffic)

Port	TCP	UDP	Protocol / Description
465	✓		SMTPS (SMTP over TLS/SSL)
514		✓	Syslog (system logging)
546		✓	DHCPv6 client
547		✓	DHCPv6 server
587	✓		SMTP (submission, STARTTLS)
636	✓	✓	LDAPS (LDAP over TLS/SSL)
853	✓	✓	DNS over TLS
873	✓		Rsync (file synchronization)
8883	✓		MQTT over TLS (IoT messaging)
9000	✓		WebSocket (HTTP upgrade)
9092	✓		Kafka (streaming platform)
9443	✓		HTTPS (management APIs)
993	✓		IMAPS (IMAP over TLS/SSL)
995	✓		POP3S (POP3 over TLS/SSL)
3306	✓		MySQL database
5432	✓		PostgreSQL database
6379	✓		Redis (key-value store)
443/UDP		✓	QUIC / HTTP/3 (secure web transport)

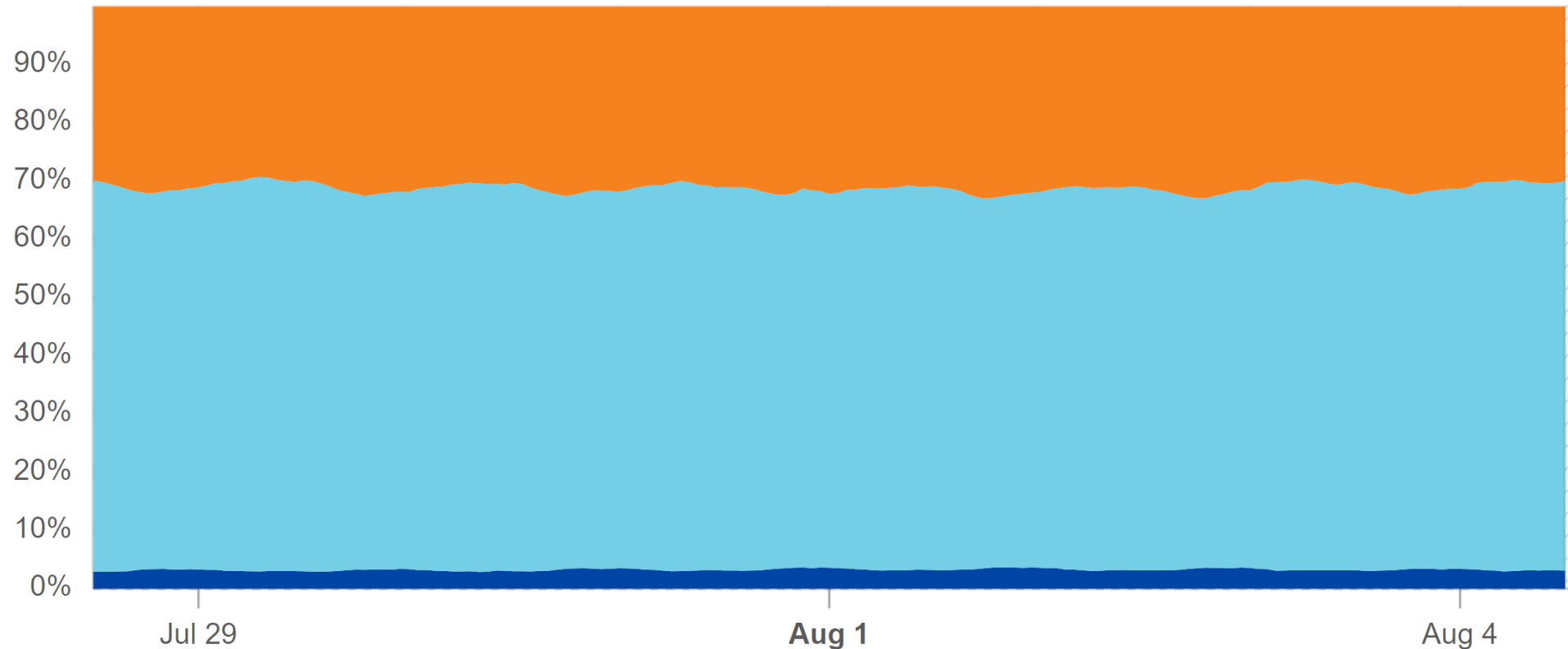
Comparison of Transport Layer Protocols

Feature	UDP	TCP	QUIC
PDU header size	8 bytes	20–60 bytes	~22–50 bytes
Connection-oriented	No	Yes	Yes
Reliable	No	Yes	Yes
Delivery	Unordered	Ordered	Both (ordered & unordered streams)
Data checksum	Optional	Yes	Yes
Checksum size	16 bits	16 bits	128 bits (AEAD-integrated)
Flow control	No	Yes	Yes
Congestion control	No	Yes	Yes
Encryption	No	Optional (via TLS/SSL)	Always (TLS 1.3 integrated)
Header compression	No	No	Yes (built-in)
0-RTT connection setup	No	No	Yes
Multiplexing streams	No	No	Yes
Used by	DNS, VoIP, streaming	HTTP/1.1, SMTP, IMAP	HTTP/3, modern web apps

Comparison of Transport Layer Protocols

- CloudFlare's distribution of secure HTTP requests by protocol (2026)

■ TLS 1.2 ■ TLS 1.3 ■ QUIC
3.2% **65.6%** **31.2%**



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Sources

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